CS 498 HW0 Report
Perspective, Homography, and Epipolar Geometry

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1 Homography and 2D insertion

We normalized the court points XY and image points UV separately by translating every point by negative mean distance and then scaling to have normalized points that have $\mu = 0$ and mean distance from the origin of $\sqrt{2}$. Then I construct the DLT by using the design matrix with

$$\begin{bmatrix} x & y & 1 & 0 & 0 & 0 & -ux & -uy & -u \\ 0 & 0 & 0 & x & y & 1 & -vx & -vy & -v \end{bmatrix} \mathbf{h} = \mathbf{0}, \qquad \mathbf{h} = \mathrm{vec}(H) \in \mathbb{R}^9$$

Since contours have 4 points in total, we can stack 4 groups of equations to derive $A \in \mathbb{R}^{2N \times 9}$ where $N = 4$. Eventually, we solve the $\mathbf{h}$ by applying SVD to $A\mathbf{h} = 0$ for non-trivial solution.

My maximum corner-transfer error is 0.00 px. The alpha-composition equation is follwoing:

$$I = \alpha\, F + (1 - \alpha)\, B$$

In this equation, F is the foreground pixel, and B is the background pixel. The foreground pixel now has 4 parameters: r, g, b, and alpha that represents the opacity level. 1 is the solid foreground pixel and 0 is the completely transparent foreground pixel. Hence, we apply this equation to every color channel of every two corresponding foreground and background pixels. The mechanism behind the equation is the weighted average of two pixel values, and the weight sum must be 1.

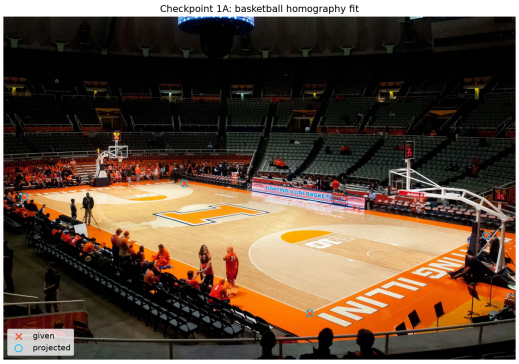


Figure 1: Basketball court homography



Figure 2: Basketball court insertion

1.1 Tennis transfer and camera height

We could derive the XY coordinates by using Figure 4 of tennis-court dimensions in the handout. The RMSE (tennis_8_anchor_reprojection_rmse_px) is approximately 3.66991. By intuition, the reason why we cannot determine a unique camera height without intrinsics is that the view from a camera closer to the object with smaller focal length is same as the one from a camera farther from the object with larger local length. Mathematically, for a camera looking straight down, the projection depends only on the ratio $\frac{f}{h}$. Hence, if f is derived from known intrinsics, height could be derived as well.



Figure 3: Tennis court alignment

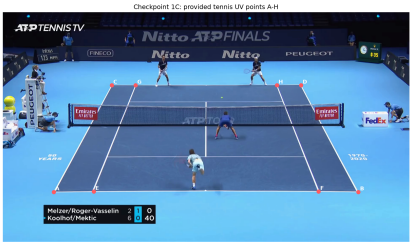


Figure 4: Court correspondences



Figure 5: Tennis court insertion

Point	Description	u (px)	v (px)	X (m)	Y (m)
A	near left outer corner	240	845	−5.485	0
B	near right outer corner	1679	845	5.485	0
C	far left outer corner	514	343	−5.485	23.77
D	far right outer corner	1408	343	5.485	23.77
E	near left inner corner	429	843	−4.115	0
F	near right inner corner	1491	843	4.115	0
G	far left inner corner	627	343	−4.115	23.77
H	far right inner corner	1294	343	4.115	23.77

Table 1: Corresponding UV and XY

2 Camera projection and 3D insertion

After normalizing the 3D points and 2D points respectively in the similar method described in section 1, we build the projection DLT as follows, where $\mathbf{X}_i$ is the coordinate of vertex of the bunny, P is camera the projection matrix, $[x_i, y_i, z_i]^T$ is the coordinate of 2D image pixel later normalized by z_i:

$$\mathbf{X}_i = \begin{bmatrix} X_i \\ Y_i \\ Z_i \\ 1 \end{bmatrix}, \qquad P = \begin{bmatrix} \mathbf{p}_1^T \\ \mathbf{p}_2^T \\ \mathbf{p}_3^T \end{bmatrix}, \qquad \begin{bmatrix} x_i \\ y_i \\ z_i \end{bmatrix} = P\mathbf{X}_i = \begin{bmatrix} \mathbf{p}_1^T\mathbf{X}_i \\ \mathbf{p}_2^T\mathbf{X}_i \\ \mathbf{p}_3^T\mathbf{X}_i \end{bmatrix}$$

Then we can get homogeneous coordinate by dividing z_i

$$u_i = \frac{x_i}{z_i} = \frac{\mathbf{p}_1^T\mathbf{X}_i}{\mathbf{p}_3^T\mathbf{X}_i}, \qquad v_i = \frac{y_i}{z_i} = \frac{\mathbf{p}_2^T\mathbf{X}_i}{\mathbf{p}_3^T\mathbf{X}_i}$$

After cross-multiplication and moving the right term to the left, we have:

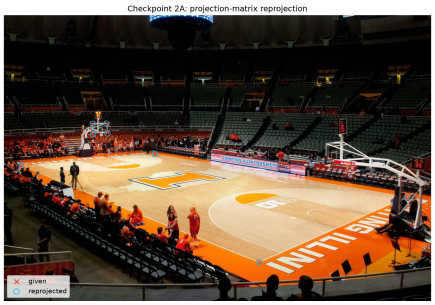


Figure 6: Reprojection



Figure 7: Bunny insertion

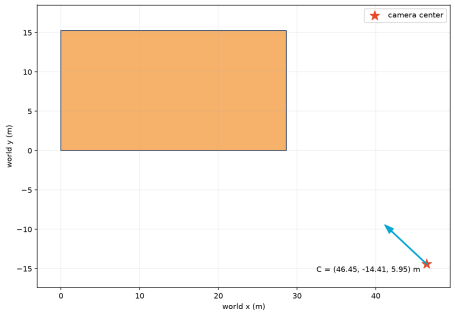


Figure 8: Camera pose

$$\mathbf{p}_1^T\mathbf{X}_i - u_i\mathbf{p}_3^T\mathbf{X}_i = 0,$$
$$\mathbf{p}_2^T\mathbf{X}_i - v_i\mathbf{p}_3^T\mathbf{X}_i = 0$$

Then we transform it to express unknowns in a vector, which is the exact form written in the code

$$\mathbf{p} = \begin{bmatrix}\mathbf{p}_1\\ \mathbf{p}_2\\ \mathbf{p}_3\end{bmatrix}, \qquad A_i = \begin{bmatrix} X_i & Y_i & Z_i & 1 & 0 & 0 & 0 & 0 & -u_iX_i & -u_iY_i & -u_iZ_i & -u_i \\ 0 & 0 & 0 & 0 & X_i & Y_i & Z_i & 1 & -v_iX_i & -v_iY_i & -v_iZ_i & -v_i \end{bmatrix}, \qquad A_i\mathbf{p} = \mathbf{0}$$

We know P is $\mathbb{R}^{3\times4}$ matrix with 12 coefficients. With the use of 8 pairs of points in total, we can stack 8 groups of equations to derive $A \in \mathbb{R}^{2N\times12}$ where $N = 8$. Eventually, we solve the $\mathbf{p}$ by applying SVD to $A\mathbf{p} = 0$ for non-trivial solution as we did in planar homography.

In general, if K is known, the metric court plane is sufficient to recover camera pose. If K is unknown, general projection DLT requires non-coplanar landmarks.

To explain it mathematically. For court points on $Z = 0$, the projection equation becomes

$$\begin{bmatrix}x\\ y\\ z\end{bmatrix} = K[\mathbf{r}_1\ \mathbf{r}_2\ \mathbf{t}]\begin{bmatrix}X\\ Y\\ 1\end{bmatrix}, \qquad u = \frac{x}{z}, \quad v = \frac{y}{z}$$

Thus, the homography matrix is

$$H = K[\mathbf{r}_1\ \mathbf{r}_2\ \mathbf{t}]$$

If K is known, we can simply use $K^{-1}H$ as well as scaling to get $\mathbf{r}_1\ \mathbf{r}_2\ \mathbf{t}$. We then use cross product to get $\mathbf{r}_3$ to recover all parameters in the camera pose.

If K is unknown, points on $Z = 0$ do not constrain the third column of P, because of the following equation. So with the underdetermined linear system, we can't obtain P, where camera pose comes from.

$$P\begin{bmatrix} X \\ Y \\ 0 \\ 1 \end{bmatrix} = X\mathbf{c}_1 + Y\mathbf{c}_2 + \mathbf{c}_4, \qquad P = [\mathbf{c}_1\ \mathbf{c}_2\ \mathbf{c}_3\ \mathbf{c}_4]$$

The mean and maximum reprojection errors are 2.75245 pixels and 5.09944 pixels, respectively. The estimated camera parameters are

$$K = \begin{bmatrix} 763.89 & 258.94 & 481.84 \\ 0 & 361.76 & -202.23 \\ 0 & 0 & 1 \end{bmatrix},$$

$$R = \begin{bmatrix} 0.71 & 0.71 & 0.02 \\ -0.6 & 0.61 & -0.51 \\ -0.37 & 0.35 & 0.86 \end{bmatrix},$$

$$t = \begin{bmatrix} -22.72 \\ 39.86 \\ 17.21 \end{bmatrix} \text{ m}, \qquad C = \begin{bmatrix} 46.45 \\ -14.41 \\ 5.95 \end{bmatrix} \text{ m}$$

Since the court lies on $Z = 0$, the optical-center height is

$$h = C_z = 5.95\,\text{m}$$

The camera center C satisfies $RC + t = 0$. Since $R^{-1} = R^\top$,

$$C = -R^\top t$$

by the property of rotation matrix Using the computed parameters,

$$\left\| C - (-R^\top t) \right\|_2 = 5.02 \times 10^{-15}\,\text{m},$$

which verifies this relationship.

3 Two-view epipolar geometry

The points in each image are normalized independently by translating every point by negative mean distance and then scaling to have normalized points that have $\mu = 0$ and mean distance from the origin of $\sqrt{2}$.

Using the normalized coordinates, let $\mathbf{x}_i = [u_i, v_i, 1]^T$ and $\mathbf{x}'_i = [u'_i, v'_i, 1]^T$. Expanding the epipolar constraint $(\mathbf{x}'_i)^T \mathbf{F} \mathbf{x}_i = 0$ gives one row of the design matrix:

$$\mathbf{A}_i = \begin{bmatrix} u'_i u_i & u'_i v_i & u'_i & v'_i u_i & v'_i v_i & v'_i & u_i & v_i & 1 \end{bmatrix}$$

We stack all correspondences into an 8×9 matrix and compute its full SVD. The last right singular vector, or the last row of $\mathbf{V}^T$ in the SVD, is reshaped row by row into a 3×3 estimate of $\mathbf{F}$.

We compute the SVD of the estimated fundamental matrix:

$$\mathbf{F} = \mathbf{U} \begin{bmatrix} \sigma_1 & 0 & 0 \\ 0 & \sigma_2 & 0 \\ 0 & 0 & \sigma_3 \end{bmatrix} \mathbf{V}^T, \qquad \sigma_1 \geq \sigma_2 \geq \sigma_3 \geq 0$$

We set the smallest singular value to zero and reconstruct to enforce rank-two matrix:

$$\mathbf{F}_2 = \mathbf{U} \begin{bmatrix} \sigma_1 & 0 & 0 \\ 0 & \sigma_2 & 0 \\ 0 & 0 & 0 \end{bmatrix} \mathbf{V}^T$$

Fundamental rank is 2, mean symmetric epipolar distance in pixels is 5.577977 px, and median symmetric epipolar distance in pixels is 4.51891 px.

Points locations might be deviated due to noise, $\mathbf{F}$ is estimated from 8 correspondences, so it cannot satisfy all the epipolar constraints of all 2000 correspondences.



Figure 9: Point correspondences

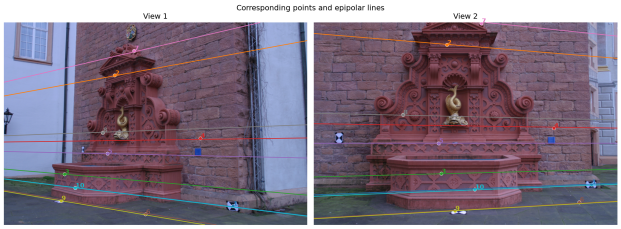


Figure 10: Epipolar lines

4 Limitations and improvements

For annotation precision, manually selected correspondences might be slightly different from the true locations, leading to error and shift in the epipolar lines, especially when only eight pairs are given. Degenerate configuration might occur when all matched points lie on the the same plane, so the correspondences might become the contours/points in the planar homography and do not uniquely determine $\mathbf{F}$. Hence, we should use correspondent points that have depth variation, and we can also annotate correspondent points multiples times to get the averaged value for better precision.

Optional foreground-preserving video bonus

Since players cover every single pixel briefly, we take the per-pixel median over 45 frames to get clean background. Then the clean background is compared with the pixel in each frame, so we can get the players when their differences are the biggest. Then we turn the RGB difference into a 0-to-1 value as a foreground mask. After projecting the logo as task 1, we use (1-mask) as the opacity value of the logo to blend foreground and background.

AI-use declaration

Check one statement.

- ☐ I did not use generative AI for this homework.
- ■ I did not use generative AI for any required part. I used it only in the optional bonus marked "Gen-AI leverage allowed." Tool/model, purpose, and affected files: `bonus_tennis_video.py`.

I confirm that generative AI did not write or complete my required code, select required correspondences, derive my required answers, generate my analysis, or write my required report.